

Unit 1 Mini Project

Enterprise Student Record and Academic Management Platform

Design Thinking & Product Planning Documentation

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Submitted for: Unit 1 – Software Product Engineering

Domain: Higher Education / Academic Institution Management

1. Introduction to Software Product Development

Modern academic institutions no longer manage student records through ad-hoc spreadsheets or isolated scripts. A production-grade Student Record and Academic Management Platform is built through a structured engineering lifecycle — spanning domain research, design thinking, requirements engineering, architecture, and phased delivery — long before a single line of application code is written.

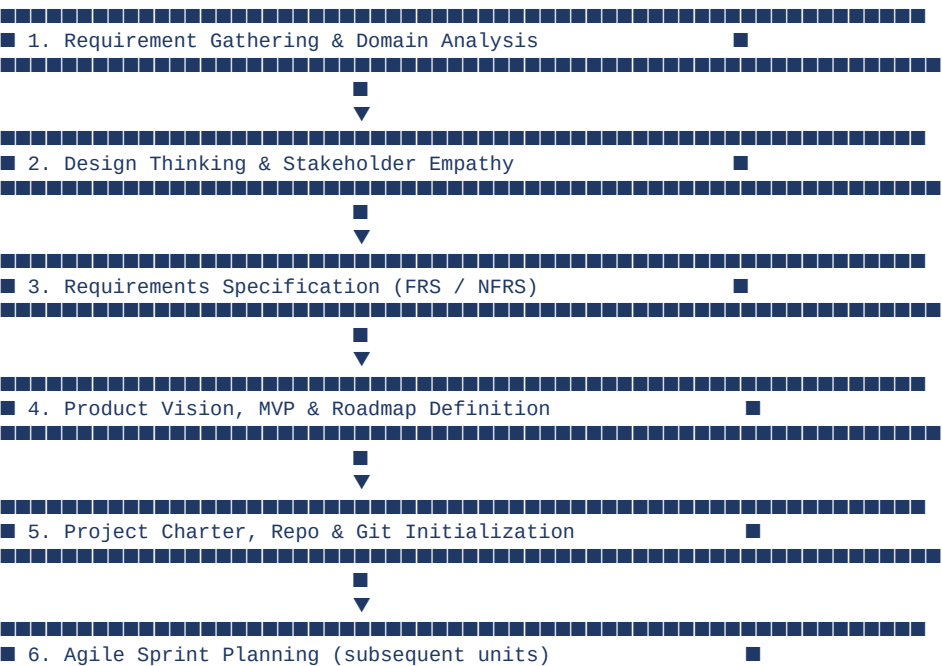
1.1 Comparative Analysis: Programming vs. Software Product Engineering

Dimension	Programming / Coding	Software Product Engineering
Primary Goal	Translating logic into executable code.	Building a scalable, maintainable, valuable academic system.
Scope	Isolated functions or single scripts.	Full lifecycle: requirements, architecture, UX, delivery, monitoring.
Success Metric	Code compiles and passes test inputs.	Faculty/student adoption, data accuracy, institutional outcomes.
Mindset	Technology-centric (“How do I write this function?”)	Product-centric (“What problem are students and faculty facing?”)

1.2 Root Causes of Academic Software Project Failures

- **Misaligned Requirements:** Building modules (e.g. elaborate analytics) that registrars or students never actually asked for.
- **Technology-Centric Bias:** Selecting frameworks for novelty rather than fit with institutional data and compliance needs.
- **Communication Silos:** Gaps between academic administrators, developers, and IT support during requirement capture.
- **Late Feedback:** Involving faculty and students only after development, making late-stage pivots costly.

System Development Lifecycle (SDLC) Followed in This Project





2. Business Understanding: Academic Record Management

Before defining any feature, the academic operations of a higher education institution must be studied end-to-end — from a student's admission to their final result declaration.

2.1 Core Operations in Academic Management

- **Admissions:** Capturing applicant data, eligibility verification, and enrolment into programs.
- **Course Registration:** Mapping students to semesters, electives, and credit-based course structures.
- **Attendance Management:** Recording daily/period-wise attendance and computing eligibility percentages.
- **Examination Management:** Scheduling exams, hall-ticket generation, and internal assessment tracking.
- **Result Management:** Grade computation, transcript generation, and academic report distribution.

2.2 Manual Process Challenges vs. Automated Platform Solutions

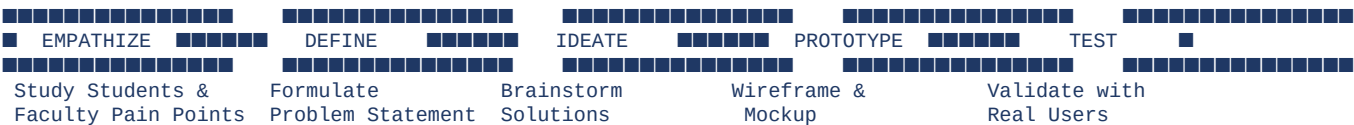
Manual Bottleneck	Automated Platform Solution	Business Outcome Metric
Paper-based admission forms and physical filing.	Digital admission & centralized student database.	Onboarding time reduced from days to minutes.
Register-based attendance marking.	Digital attendance capture with real-time percentage tracking.	Attendance errors reduced by over 90%.
Manual mark-sheet tabulation.	Automated grade computation & transcript generation engine.	Result processing time cut from weeks to hours.
Scattered records across departments.	Unified, role-based academic record repository.	Record retrieval time reduced from 20 min to seconds.

3. Design Thinking Methodology

Design Thinking is applied as an iterative, human-centred framework that prioritises empathy for students, faculty, and administrators before technology decisions are made.

- **Human-Centred Innovation:** Designing around how students and faculty actually work, not how the system is easiest to build.
- **Creative Ideation:** Exploring a wide solution space before narrowing to a technical design.
- **Iterative Prototyping:** Validating wireframes cheaply before backend development begins.

Applying the Five Stages of Design Thinking



Stage 1 – Empathize

Stakeholders observed in their natural academic workflow include: Students, Faculty Members, Academic Administrators, Examination Cell Staff, and IT Support Personnel.

Stage 2 – Define

“Students and faculty face recurring delays and errors in registration, attendance, and result processing due to fragmented manual record-keeping. Administrators lack a centralized, real-time view of academic data, leading to reporting errors, compliance risk, and poor stakeholder satisfaction.”

Stage 3 – Ideate

- Self-service student portal for registration and result viewing.
- Digital attendance capture with automated shortage alerts.
- Role-based dashboards for faculty, examination cell, and administrators.
- Automated transcript and report-card generation.

Stage 4 – Prototype

Low-fidelity wireframes planned for core UI modules: Student Login & Dashboard, Course Registration Screen, Faculty Attendance Marking Screen, and Administrator Reporting Panel.

Stage 5 – Test

Paper/wireframe prototypes are to be reviewed with sample students and faculty to validate navigation flow and task completion speed before frontend engineering begins.

4. Enterprise Expansion: IBM Design Thinking

IBM Design Thinking scales human-centred design for institution-wide delivery through three pillars:

Focus on User Outcomes

Feature Goal	Outcome Focus
“Build a student attendance table.”	“Enable faculty to mark a class's attendance in under 60 seconds.”

Restless Reinvention

Treating continuous deployment as an ongoing refinement cycle — regularly analysing usage data from registrars and students to improve workflows and response speed.

Diverse Empowered Teams

Cross-functional collaboration across Product Owners, UX Designers, Software Engineers, QA, Database Administrators, and DevOps Engineers.

The IBM Loop: Observe → Reflect → Make





5. Stakeholder User Personas

Persona	Primary Objectives	Key Frustrations	Target Platform Feature
Ananya (Student)	Register for courses fast; track attendance and results.	Long registration queues; delayed result declaration.	Self-service portal; real-time attendance & result view.
Prof. Mehta (Faculty)	Mark attendance quickly; upload internal marks accurately.	Manual attendance sheets; error-prone mark entry.	Digital attendance & gradebook module.
Dr. Kapoor (Academic Administrator)	Monitor institution-wide academic compliance and reporting.	No real-time visibility; manual report compilation takes days.	Centralized analytics dashboard with report export.
Examination Cell Staff	Schedule exams and process results without errors.	Manual tabulation errors; tight result-declaration deadlines.	Automated grade computation & transcript engine.

6. Stakeholder Responsibility Matrix

Stakeholder Group	Key Roles	Responsibilities & Project Expectations
Business Stakeholders	College Management, Registrar, Students, Faculty	Define business vision, validate usability, ensure academic compliance.
Engineering Stakeholders	Software Developers, UI/UX Designers, QA Engineers	Architect the codebase, build responsive interfaces, write and run tests.
Operations Stakeholders	DevOps Engineers, Database Administrators, IT Support	Provision infrastructure, maintain data integrity, automate deployment pipelines.

7. Requirements Engineering Specification

7.1 Functional Requirements (FRS)

- **FR-01 Authentication:** Role-Based Access Control (RBAC) for Admin, Faculty, Student, and Examination Cell.
- **FR-02 Student Registration:** Full CRUD for student profiles, programs, and semester enrolment.
- **FR-03 Attendance Management:** Period-wise attendance capture with automated percentage computation.
- **FR-04 Examination & Result Engine:** Mark entry, grade computation, and transcript generation.
- **FR-05 Reporting:** Configurable academic and compliance report generation for administrators.

7.2 Non-Functional Requirements (NFRS)

- **NFR-01 Security:** Password hashing, session/token management, and HTTPS encryption for student data.

- **NFR-02 Performance:** Sub-second response for search and attendance queries under concurrent load.
- **NFR-03 Reliability & Scalability:** High-availability deployment able to scale during admissions/exam peaks.
- **NFR-04 Usability:** Accessible, intuitive interfaces for non-technical faculty and administrative staff.

8. Product Vision & Minimum Viable Product (MVP)

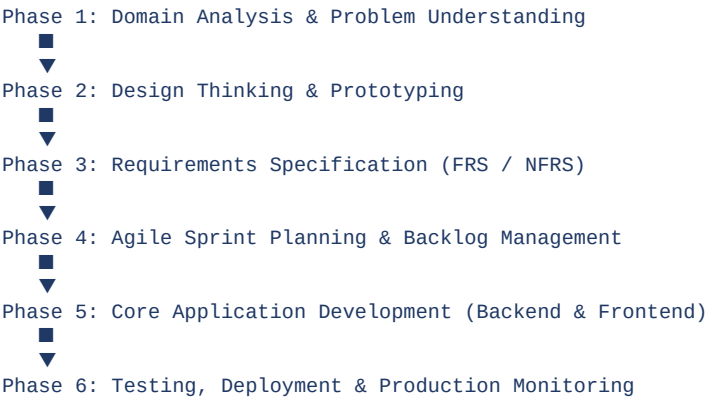
“To deliver a unified, secure, and intuitive Student Record and Academic Management Platform that eliminates manual administrative overhead and gives students, faculty, and administrators real-time visibility into academic data.”

MVP Release Scope (Sprint 1–4)

To maintain agility, non-essential capabilities (e.g. AI-based performance prediction, alumni modules) are deferred to later phases. The MVP includes:

- Student & Faculty Login with Role-Based Access
- Student Registration & Record Management
- Attendance Marking & Percentage Tracking
- Basic Result Entry & Academic Report Generation

9. Enterprise Engineering Roadmap



10. Project Repository Structure

```
student-academic-management-platform/
├── .github/           # CI/CD Workflows
├── app/               # Application Source Code
│   ├── modules/      # Business Logic (Students, Attendance, Results)
│   ├── database/     # Schemas, Migrations & ORM Models
│   └── tests/         # Automated Unit & Integration Tests
├── docs/              # Architecture Diagrams & FRS/NFRS Specs
├── docker/            # Dockerfiles & Compose Setup
├── monitoring/        # Logging & Monitoring Configs
└── README.md          # Project Onboarding & Setup Instructions
```

11. Deliverables Checklist

- Business Problem Statement & Business Understanding Report
- Design Thinking Report (Empathize, Define, Ideate, Prototype, Test)
- IBM Design Thinking & IBM Loop Documentation
- Stakeholder Analysis Report & User Persona Profiles
- Functional (FRS) and Non-Functional (NFRS) Requirements Specification
- Product Vision Statement & MVP Definition
- Product Roadmap
- Initial System Architecture Diagram
- Project Charter & Repository Folder Structure